

什么是MSP430

❑ 超低功耗结构

- 0.1uA RAM 数据保持
- 0.8uA 实时时钟模式
- 250uA / 1MIPS 活动模式
- 1.8 - 3.6V 工作电压
- <6us 时钟启动
- <50nA 端口漏电流
- 采用矢量中断
- -40 - +85C 工业级温度

流行的RISC CPU

- 具有增强的性能
- 缩短代码大小

□ 高性能的模拟结构

- ### ■ 测量精密

□ 典型的应用

- 能量表
- 便携式仪器
- 智能传感

MSP430

超低功耗的单片机



Ultra-Low-Power Activity Profile



Experience the ultimate ultra-low-power MCU solution. The MSP430 from Texas Instruments exceeds today's low-power MCUs with ultra-low-power consumption and a flexible **instant-on clock system** that switches from standby to active mode in less than 1 µs. Find the perfect MCU for your low-power design needs by selecting from 11 most MSP430 product families ranging from 16 to 48 MCUs to MCUs with 80 KB flash and integrated high-performance analog and digital periphery. Get your design started today with the only tool that streamlines tool flow!

► Features

- Ultra-low-power consumption at 100 nA
- 0.1-µs RAM retention
- 1.5-µs standby-to-active start time
- 150-µs active mode exit delay (MSP430F1611)
- 1-µs wake-up time
- Master/Slave RST/CPU enables user applications as a function of the voltage
- Designed for high-performance analog peripherals ideal for precision data conversion

Learn how to take a bite out of power on "4-30 Day"

Want to know if the MSP430 is right for your next low-power application? Register to attend **FET Tech and Demo** sessions at www.ti.com/430day on April 25, 2003. Register today at www.ti.com/430day



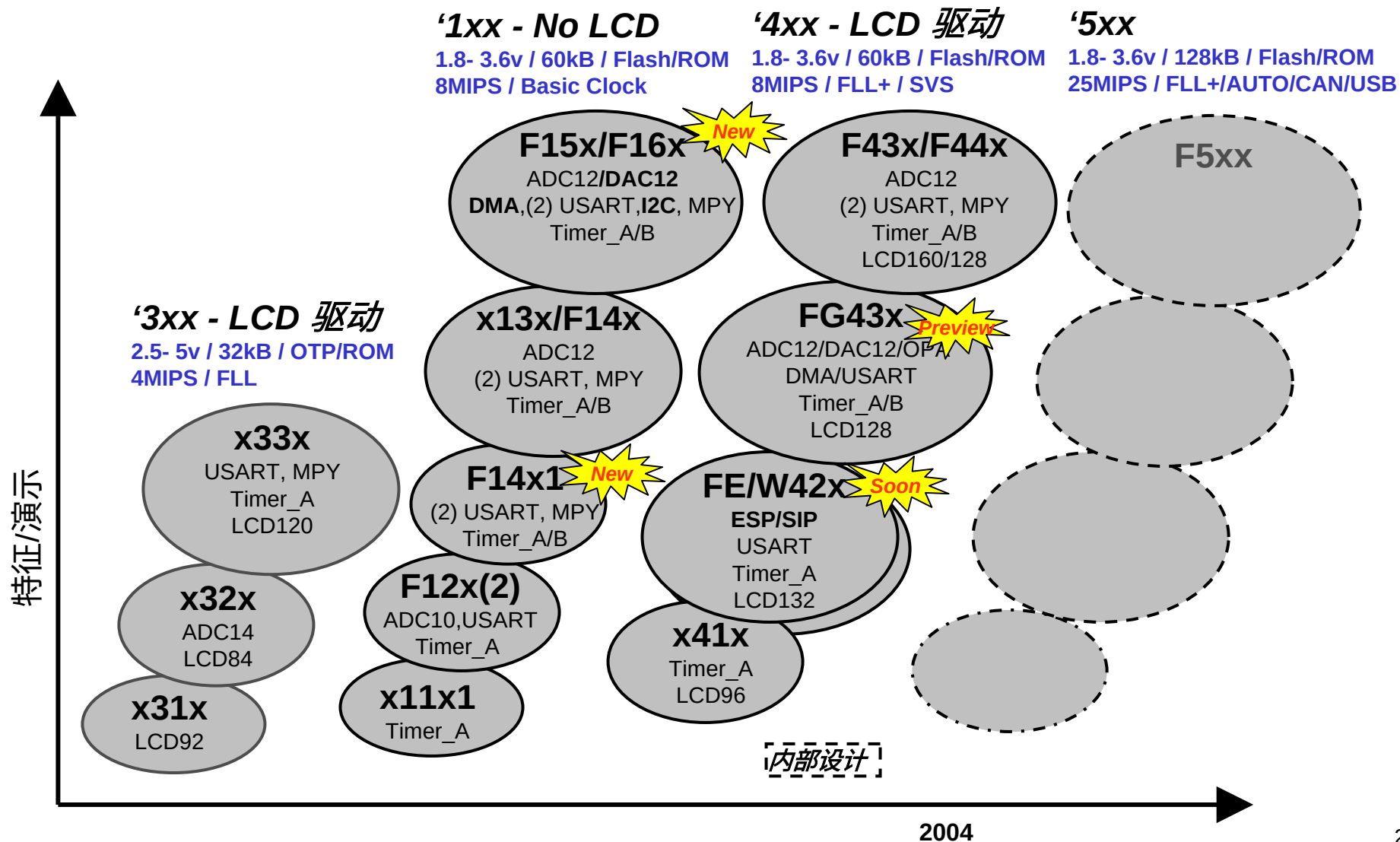
► www.ti.com/430day
1-800-477-8924, ext. 5098
 Development Tools,
 Evaluation Modules (EVMs),
 Demos and more! Register today!



REAL WORLD SIGNAL PROCESSING™

TEXAS INSTRUMENTS

MSP430 图示

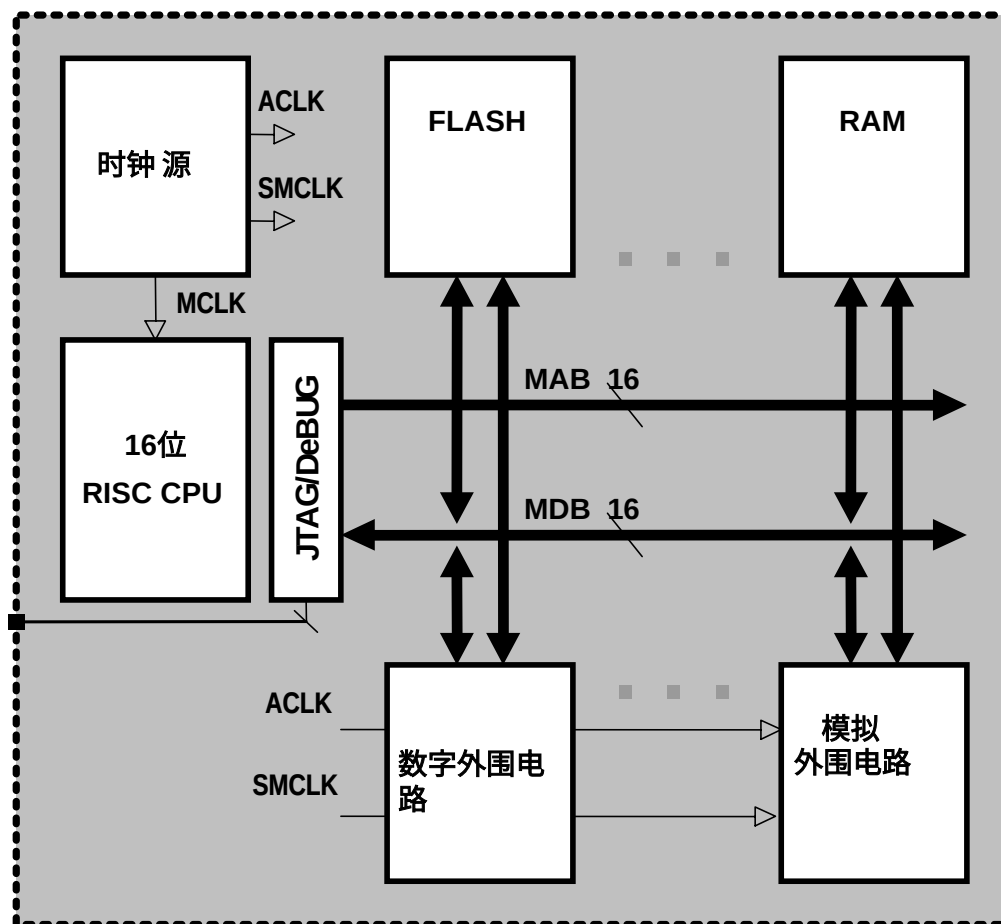


Modular von-Neumann Architecture

模块—纽曼结构

16位总线连接
CPU连接到存储器
器和到外围电路

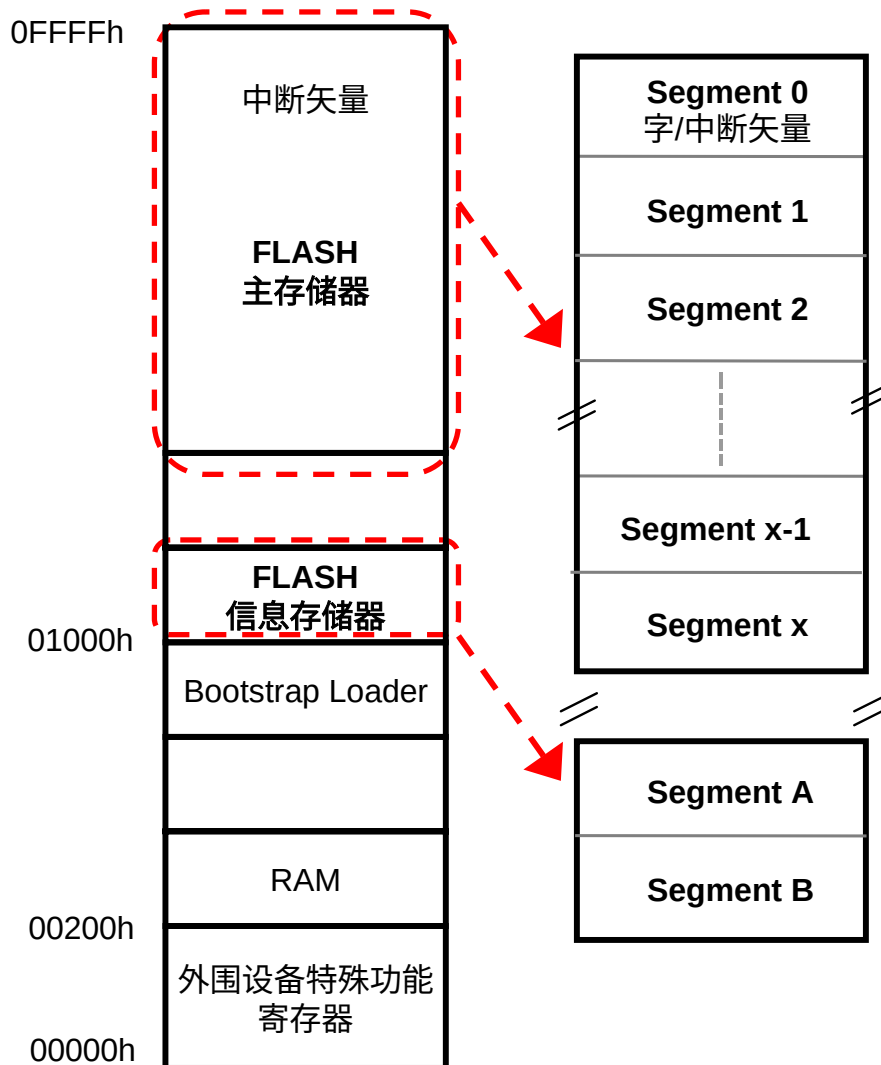
直接嵌入仿真处理,
具有JTAG接口



该结构能降低功耗,降低噪声对存储器存取的影响

16位总线宽度,数据处理更有效

Flash 存储器



- ❑ 程序、数据、外围模块统一使用 64kB的存储空间
- ❑ 整个存储空间使用相同的指令，不分页
- ❑ Flash编程可以使用位、字节、字操作；可以通过JTAG, BSL or ISP进行编程
 - 主存储器： 512B /段
 - 信息存储器： 128B/段，共2段
 - 1.8 - 3.6v 工作电压
 - 2.7 - 3.6v 编程电压
 - 100k 的擦除/编程周期
 - 数据保持 100 年
 - 60kB 空间编程时间< 5s

超低功耗Flash在线编程

Flash-timing-generator frequency, $f_{(FTG)}$		MSP430F43x, MSP430F44x	257	476	kHz
---	--	---------------------------	-----	-----	-----

$I_{(DD-PGM)}$	F-versions only (see Note 4)	Current from DV_{CC} when programming is active	$V_{CC} = 2.7\text{ V}/3.6\text{ V}$	3	5	mA
$I_{(DD-Erase)}$		Current from DV_{CC} when erase is active	$V_{CC} = 2.7\text{ V}/3.6\text{ V}$	3	5	mA
$t_{(retention)}$	F-versions only	Write/erase cycles		10^4	10^5	cycles
		Data retention $T_J = 25^{\circ}\text{C}$		100		years

4. $f_{(TCK)}$ may be restricted to meet the timing requirements of the module selected. Duration of the program/erase cycle is determined by $f_{(FTG)}$ applied to the flash timing controller. It can be calculated as follows:

$$t_{(\text{word write})} = 35 \times 1/f_{(FTG)}$$

$$t_{(\text{block write, byte 0})} = 30 \times 1/f_{(FTG)}$$

$$t_{(\text{block write, byte 1 - 63})} = 20 \times 1/f_{(FTG)}$$

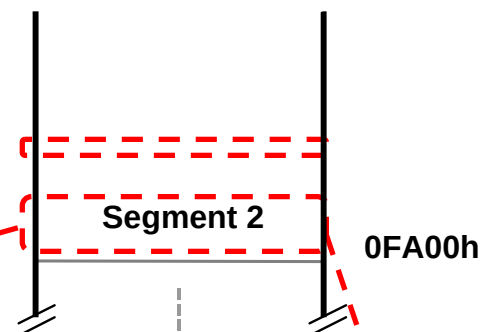
$$t_{(\text{block write, sequence end})} = 6 \times 1/f_{(FTG)}$$

$$t_{(\text{mass erase})} = 5297 \times 1/f_{(FTG)}$$

$$t_{(\text{segment erase})} = 4819 \times 1/f_{(FTG)}$$

编程 01234h @0FC00h

1. 设置寄存器
2. 写入Flash
3. 在写入期间, PC处于保持状态



```

ORG      0FA00h          ;
mov.w    #0A500h,&FCTL3   ; Lock = 0
mov.w    #0A500h+WRT,&FCTL1 ; Write bit = 1
mov.w    #01234h,&0FC00h   ; 01234h -> 0FC00h
mov.w    #0A500h+LOCK,&FCTL3 ; Lock = 1
    
```

现代的RISC/CISC MCU结构

❑ 三种指令格式

源操作数，目的操作数

目的操作数

跳转指令

❑ 51条指令

27条基本指令

24 emulated instructions

24条仿真指令

⇒ RISC(精简指令)

⇒ CISC(宏指令)

❑ 位，字节和字的处理

❑ 7种寻址模式；

寄存器寻址

索引寻址

符号寻址

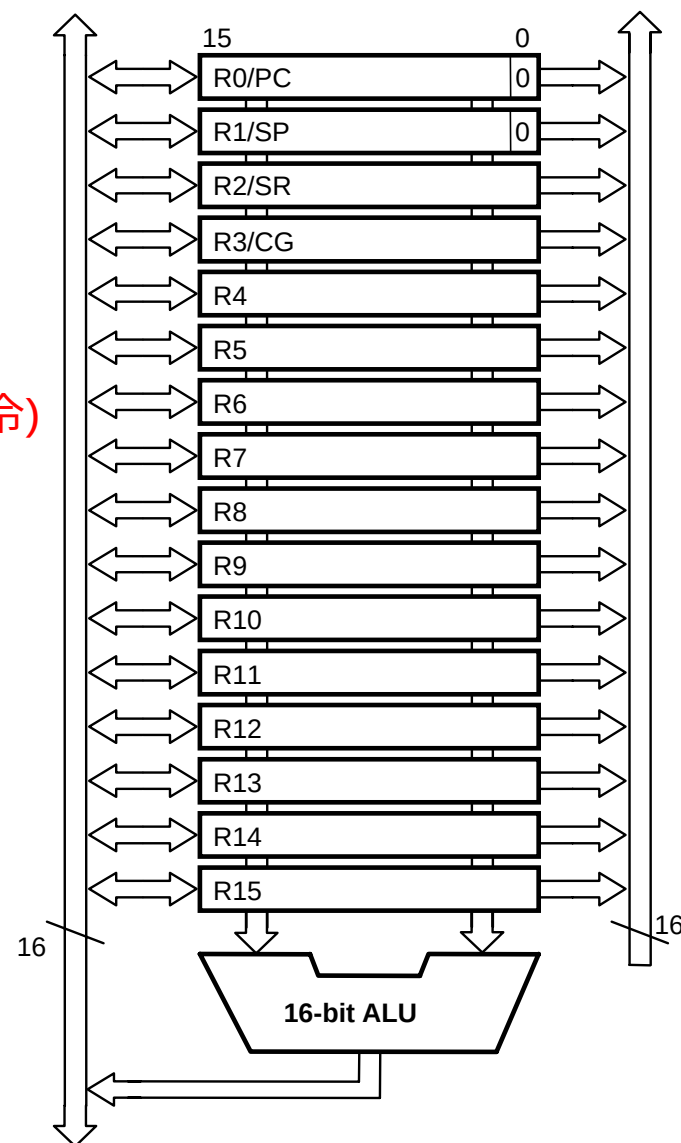
绝对寻址

直接寻址

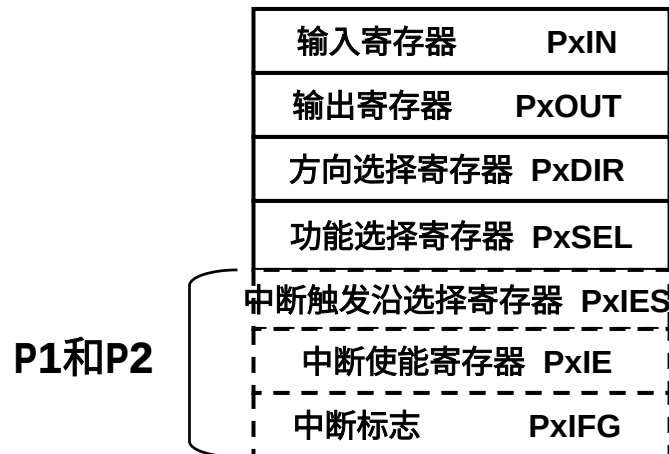
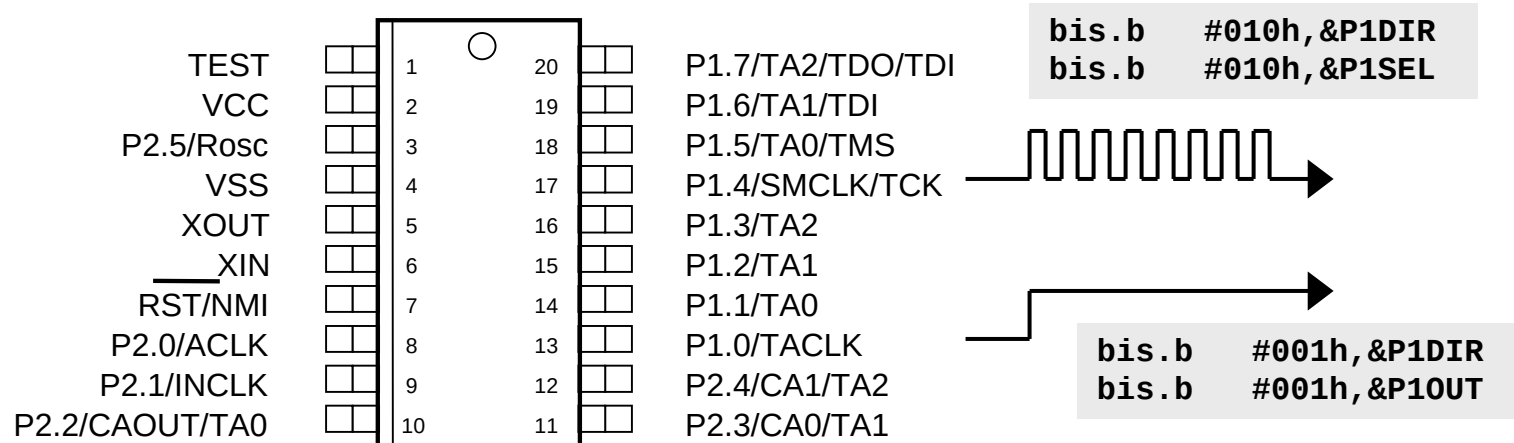
间接增量寻址

立即数寻址

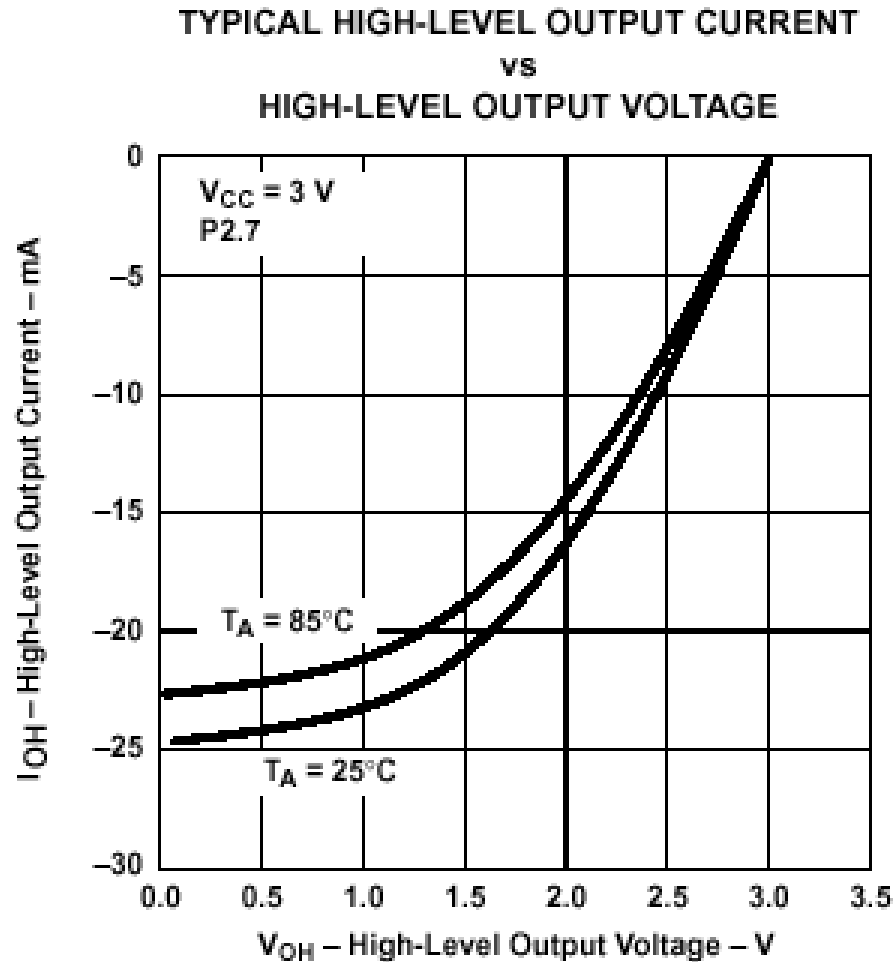
源操作数，目的操作数



输入/输出端口



端口驱动数据特性



每个端口的输出晶体管的**RDSON**能够限制电流，电流可以是灌电流和拉电流

MSP430 很容易使用

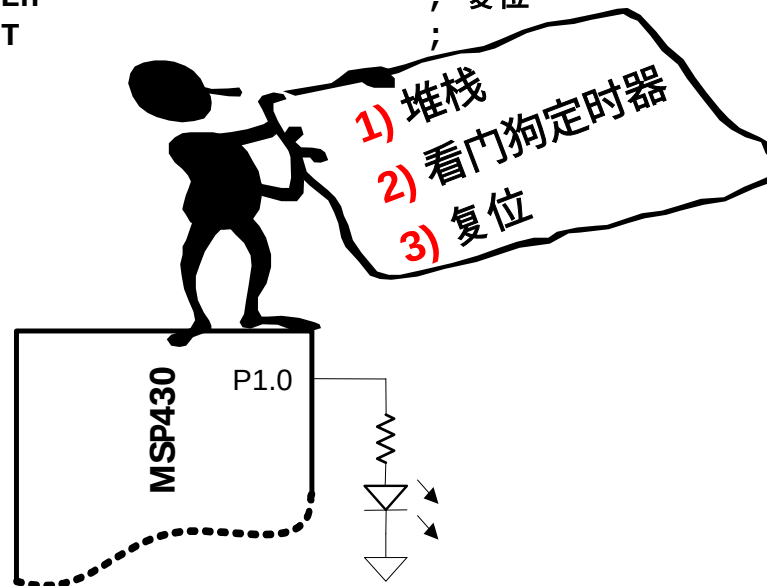
```
#include "msp430x12x2.h"
```

```

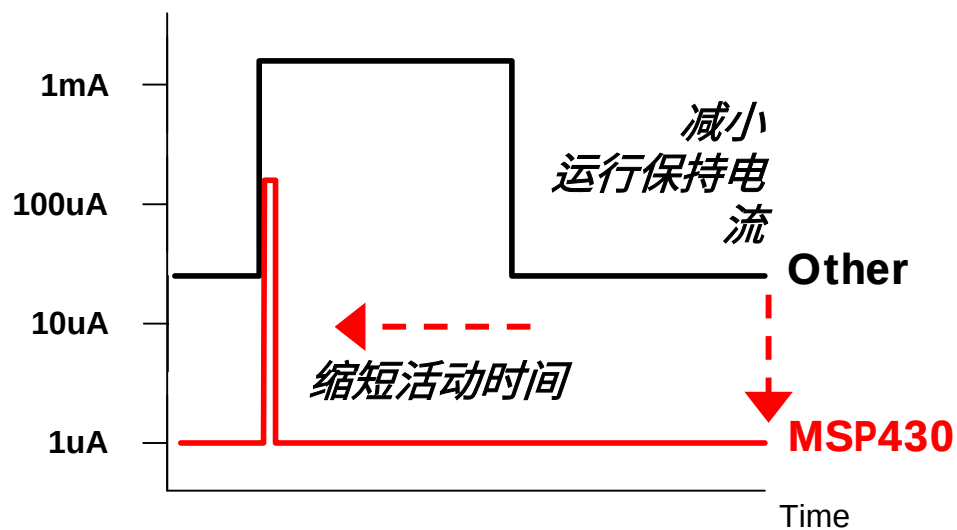
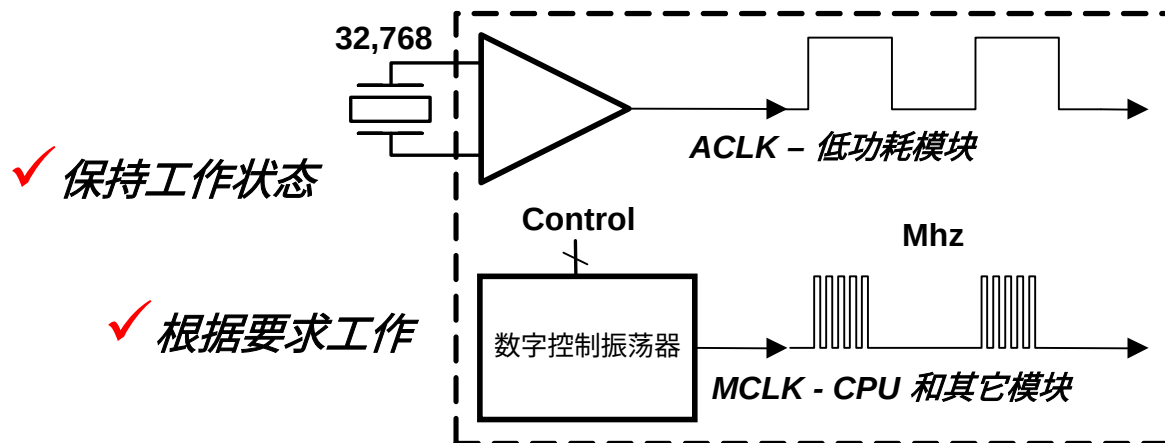
                ORG      0F000h          ; 程序开始
RESET          mov.w    #300h, SP        ; 堆栈指针
StopWDT        mov.w    #WDTPW+WDTHOLD, &WDTCTL ; 停止看门狗
SetupP1        bis.b    #01h, &P1DIR    ; P1.0 输出

Mainloop       xor.b    #01h, &P1OUT     ; P1.0 触发翻转
                mov.w    #050000, R15    ; 延时
Delay          dec.w    R15              ;
                jnz      Delay           ;
                jmp      Mainloop        ;

                ORG      0FFFEh          ; 复位
                DW       RESET          ;
                END
```

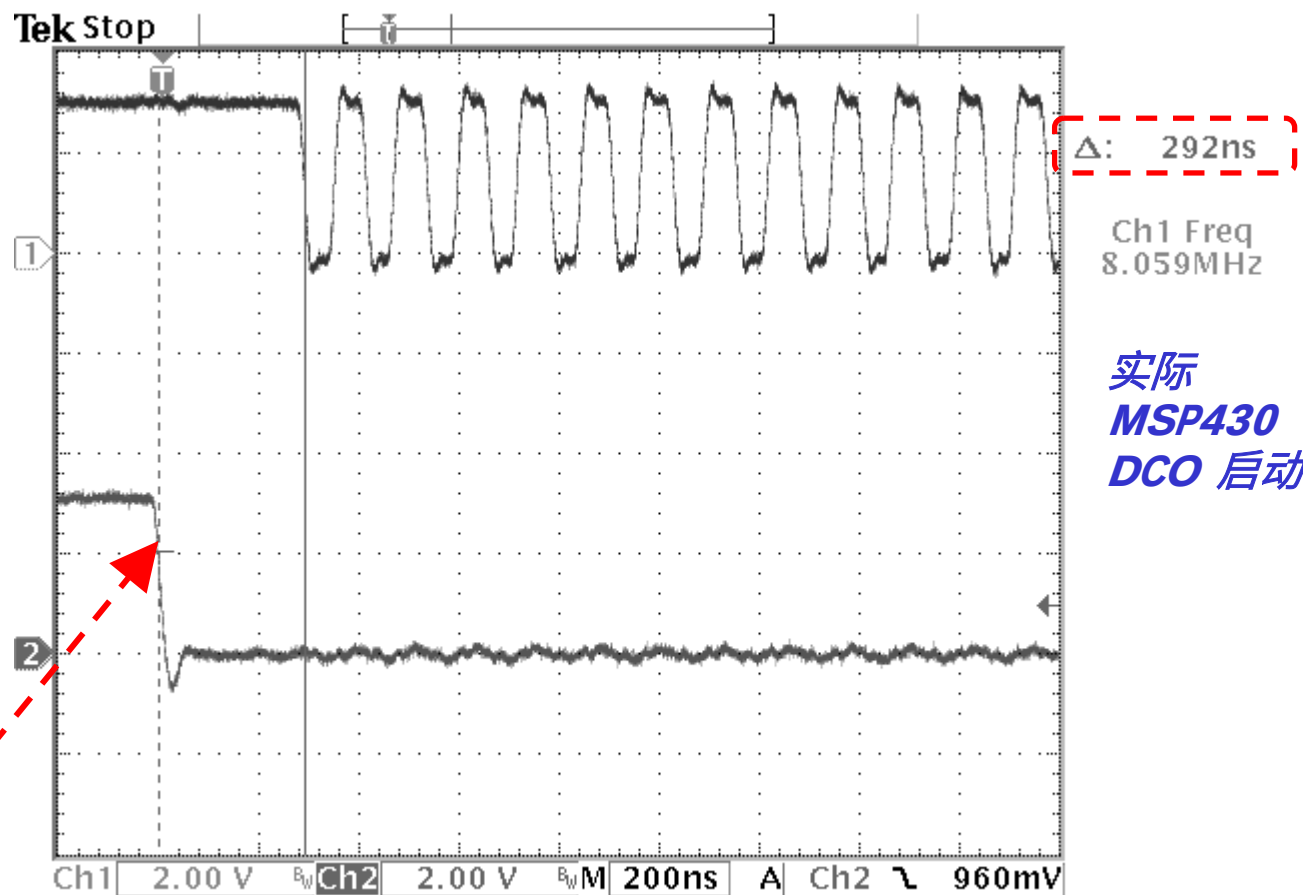


超低功耗结构



“瞬时开”不需要中间步骤

“突发工作” = 0 -到- 8MHz -稳定时间 $\ll 6\mu s$

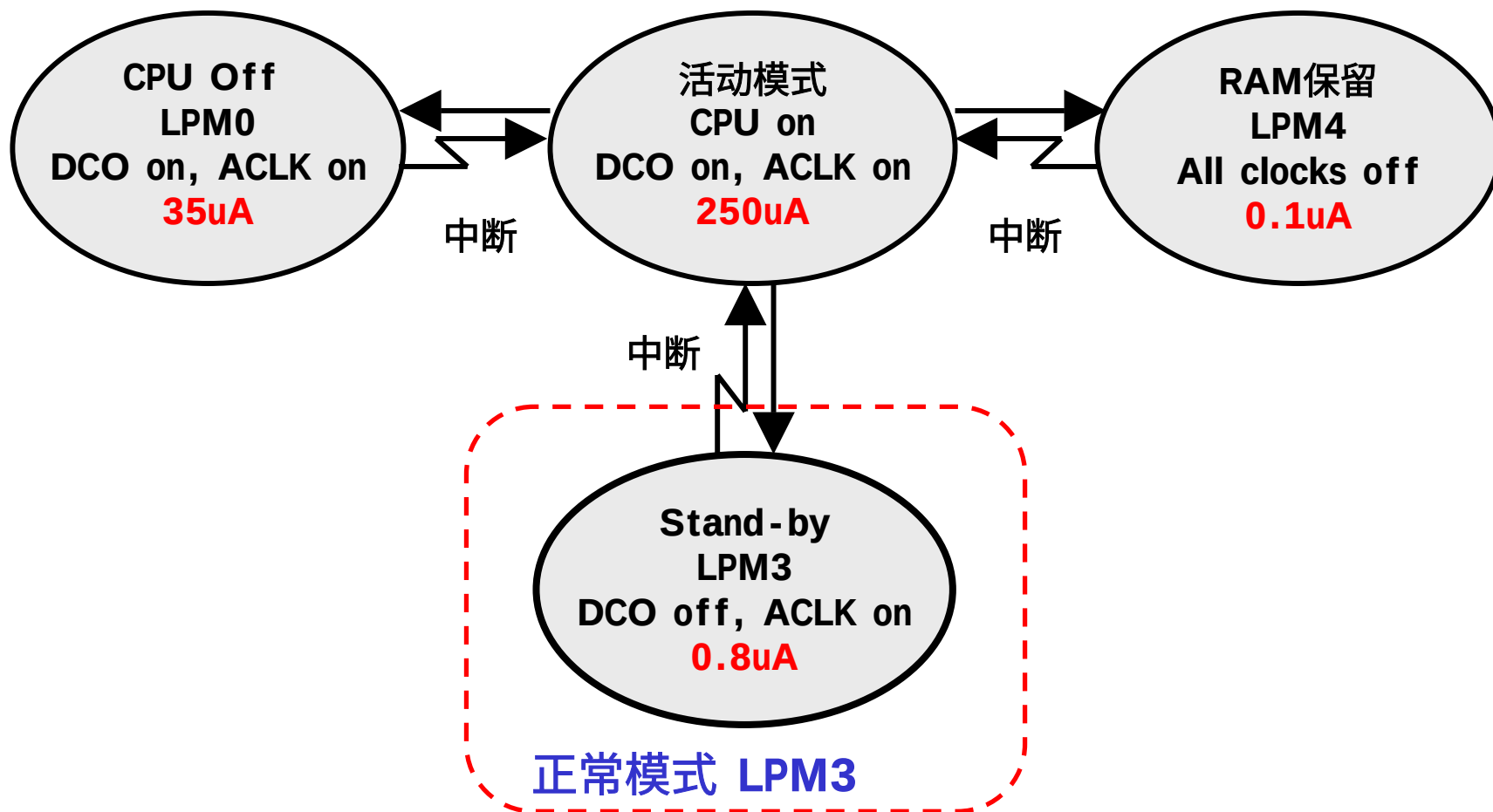


实际
MSP430
DCO 启动

中断

超低功耗事件驱动工作

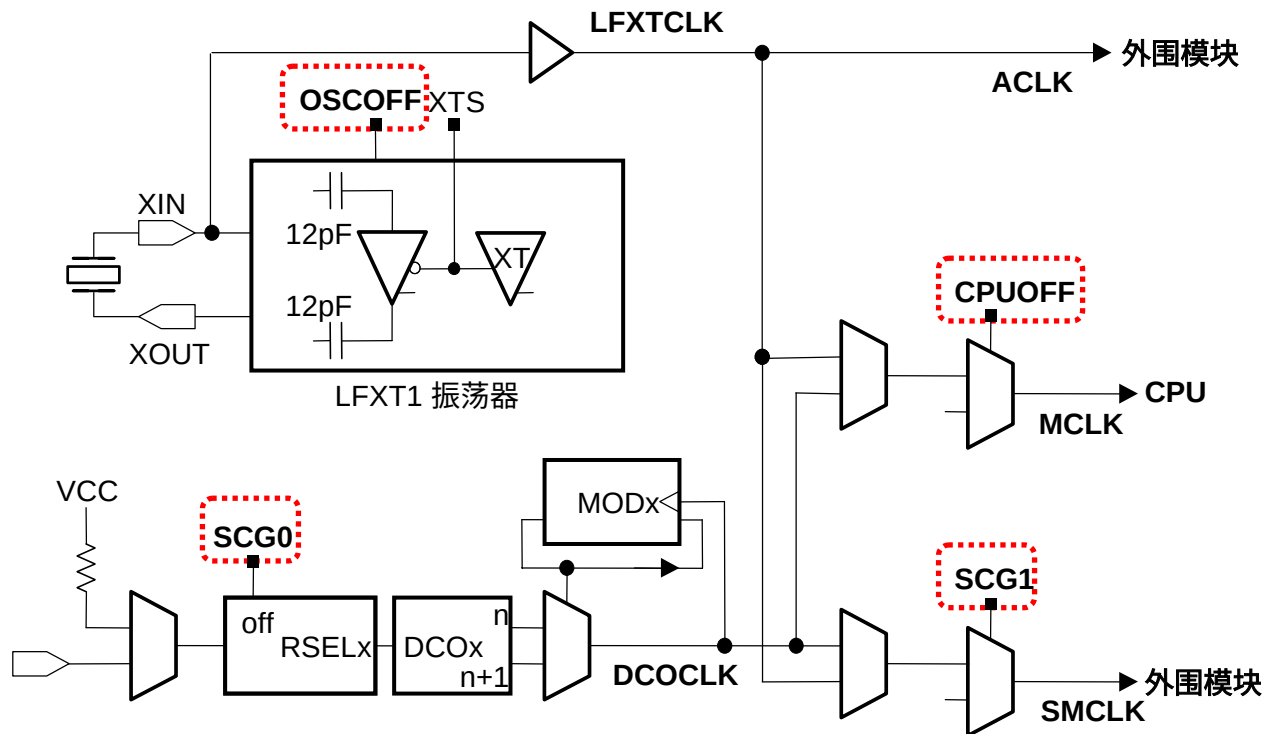
16位运行



MSP430x11x/12x 基础时钟

Reserved for MEMX	V	SCG1	SCG0	OSC OFF	CPU OFF	GIE	N	Z	C
-------------------	---	------	------	---------	---------	-----	---	---	---

R2/SR

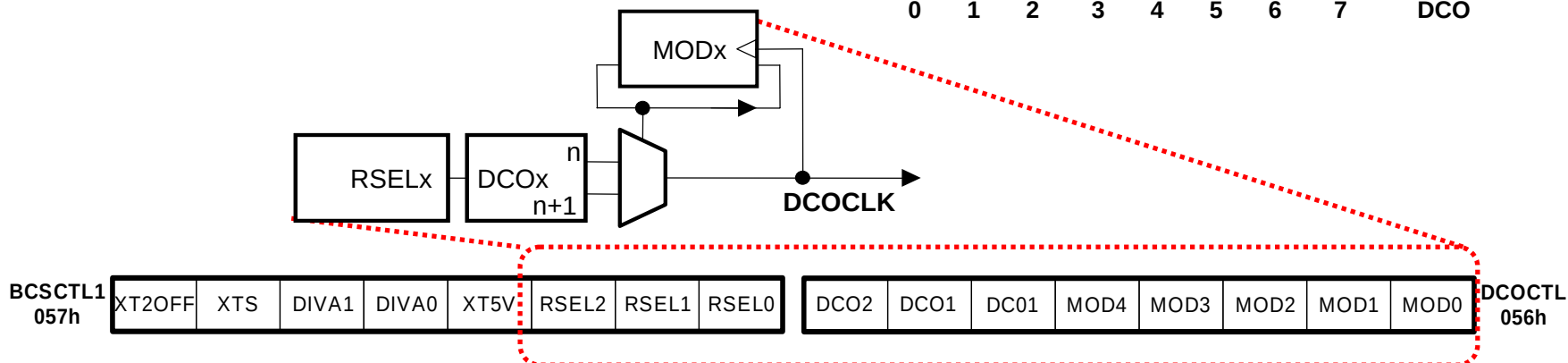
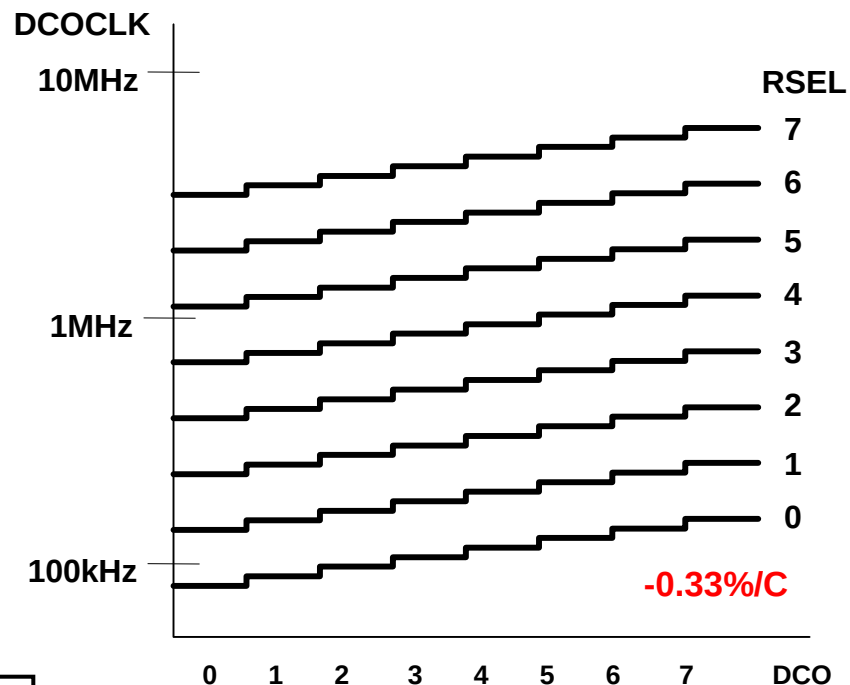


MSP430x1xx 基础时钟的频率控制

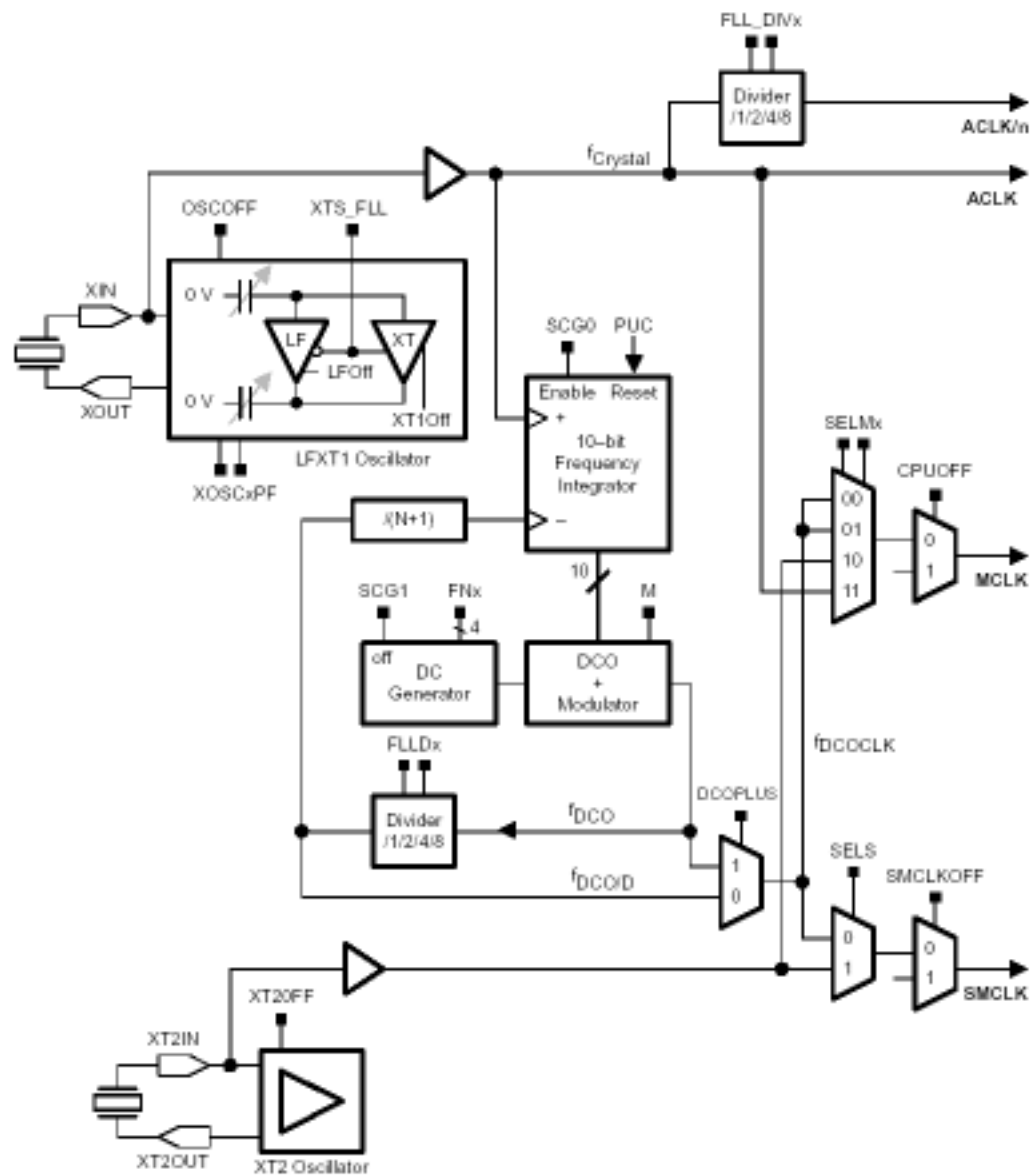
Set_2MHz `bis.b    #06h,&BCSCTL1`

Faster `inc.b    &DCOCTL`

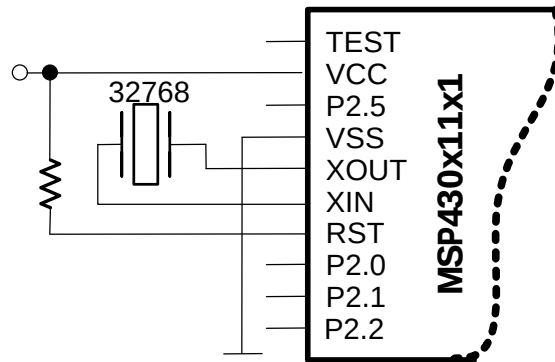
Slower `dec.b    &DCOCTL`



MSP430x43x/44x FLL+ 时钟



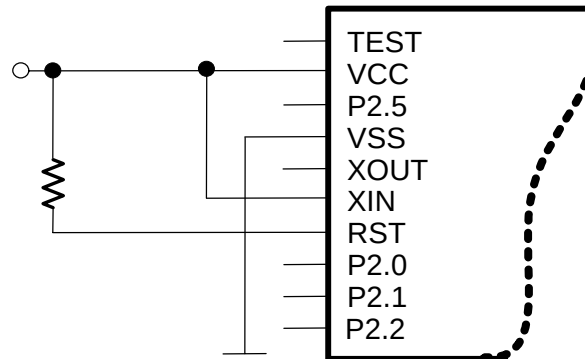
MSP430x11x/12x 基础时钟设置



ACLK = LFXT1 = 32768

MCLK = DCOCLK or LFXT1

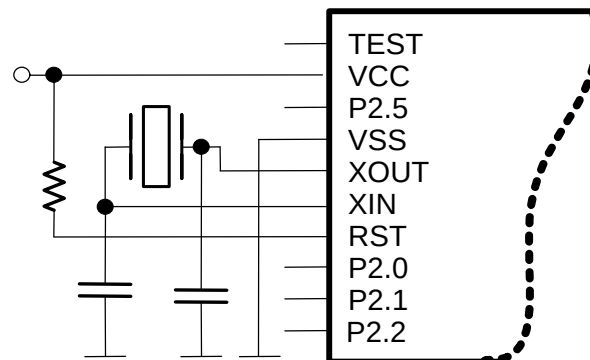
SMCLK = DCOCLK or LFXT1



ACLK = 0

MCLK = DCOCLK

SMCLK = DCOCLK

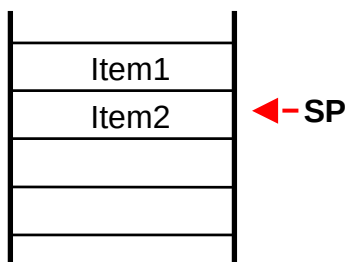


ACLK = LFXT1 = XTAL

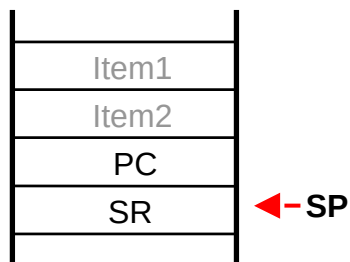
MCLK = DCOCLK or LFXT1

SMCLK = DCOCLK or LFXT1

中断处理

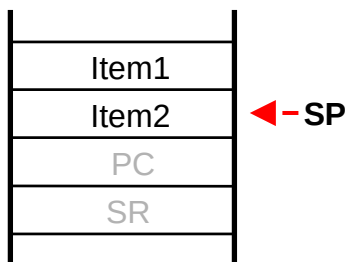


中断服务程序优先



硬件自动中断服务程序

- PC 入栈
- SR 入栈
- 中断矢量赋给 PC
- GIE, CPUOFF, OSCOFF 和 SCG1 清除
- IFG 标志位清除(单中断源标志)

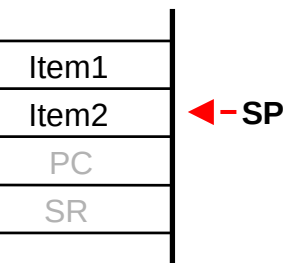
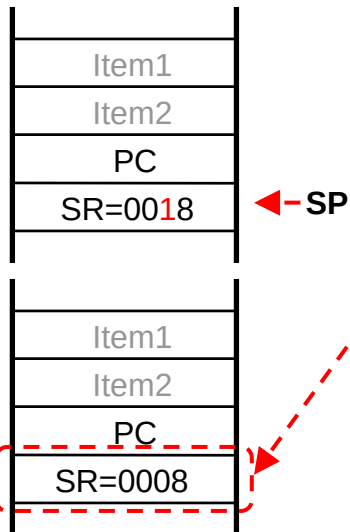
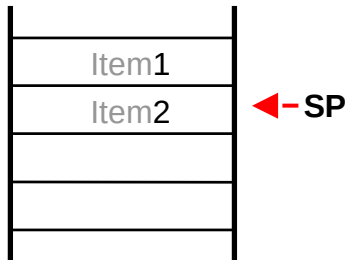


、**reti**

- SR 出栈 – 恢复原来的标志
- PC 出栈

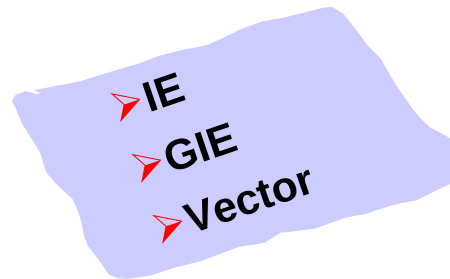
SR 和 PC 的内容自动保存在 TOS - ISR 需要11个指令周期

超低功耗模式



```
#include "msp430x11x1.h"

;
;
;
RESET   mov.w   #300h, SP
SetupWDT mov.w   #WDT_MDLY_32, &WDTCTL ; 32ms 中断
        bis.b   #WDTIE, &IE1          ; 看门狗定时器中断
        bis.b   #01h, &P1DIR          ; P1.0 输出
;
;
Mainloop bis.w   #CPUOFF+GIE, SR       ; w/ interrupts
        xor.b   #01h, &P1OUT          ; P1.0 出发
        jmp     Mainloop
;
;
WDT_ISR  bic.w   #CPUOFF, 0(SP)        ; Clear CPUOFF TOS
        reti
;
;
ORG      0FFFEh
DW        RESET
ORG      0FFF4h
DW        WDT_ISR
```



超低功耗嵌入式实时时钟

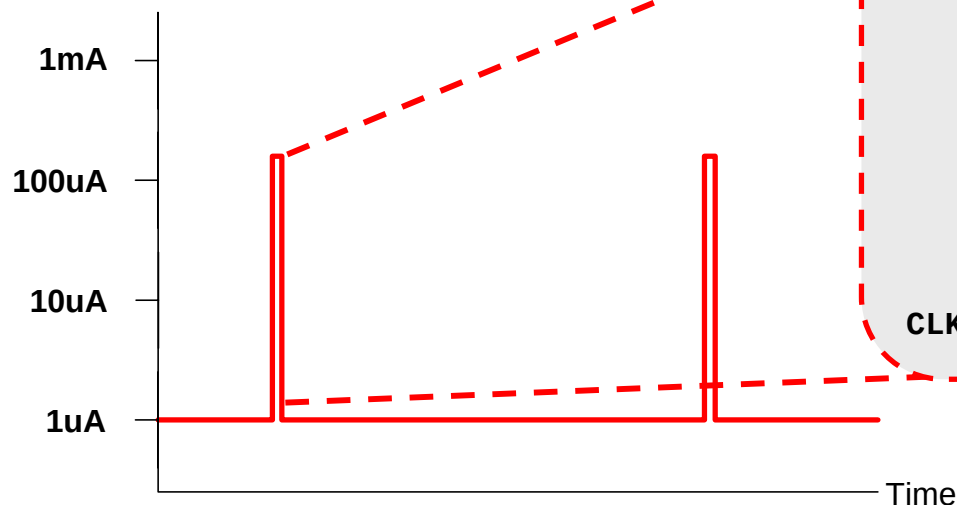
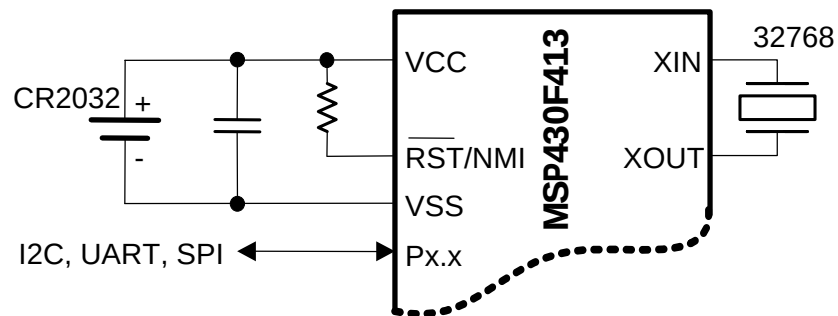
= 平均电流

= LPM3 + RTC_update

= $0.80\mu\text{A} + 250\mu\text{A} * \frac{100\mu\text{s}}{1000000\mu\text{s}}$

= $0.80\mu\text{A} + 0.030\mu\text{A}$

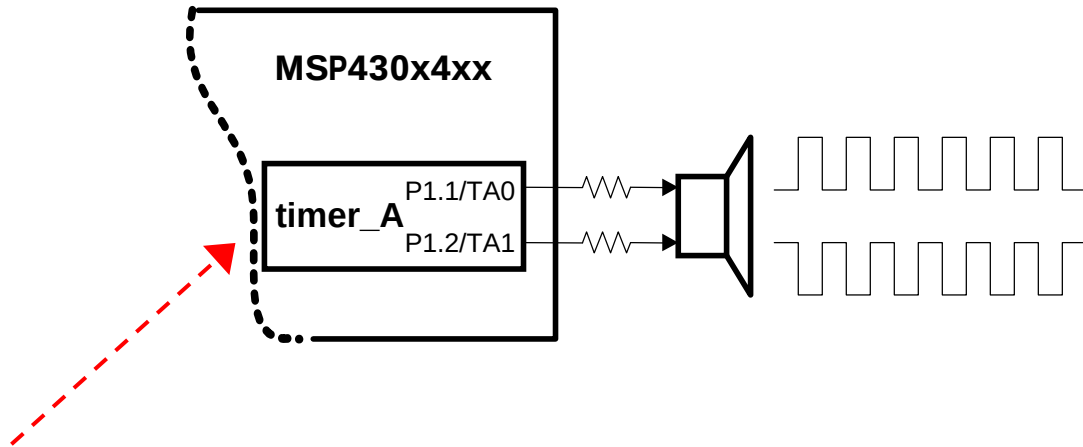
= **0.83** μA



```

; 部分代码 - 软件 RTC
setc                               ; 1 sec
dad.c.b SEC                       ; Inc sec
cmp.b #060h, SEC                  ; 1 min?
jlo CLKend                        ;
clr.b SEC                         ;
dad.c.b MIN                       ;
cmp.b #060h, MIN                  ; 60 min?
jlo CLKend                        ;
clr.b MIN                         ;
CLKend;
    
```

内部模块的正确使用



用 timer_A PWM 硬件驱动蜂鸣器不需要CPU资源

```
;  
; A bad way to toggle P1.1/2  
; All software  
Toggle    xor.b    #003h,&P1OUT  
           mov.w    #165,R4  
L1         dec.w    R4  
           jnz      L1  
           jmp      Toggle
```

低功耗应用的原则

- ❑ 最大化LPM3的时间.用32-kHz晶振作为ACLK的时钟，DCO用于CPU激活后的突发短暂运行
- ❑ 用接口模块代替软件驱动功能
- ❑ 用中断控制程序运行.
- ❑ 用可计算的分支代替标志位测试产生的分支.
- ❑ 用快速查表代替冗长的软件计算。
- ❑ 在冗长的软件计算中使用单周期的CPU寄存器。
- ❑ 避免频繁的子程序和函数调用
- ❑ 尽可能直接用电池供电
- ❑ 对外围电路进行电源管理.

超低功耗的挑战

你的系统运行在 2V 吗?

No Watt Technology!

- 150uA Run, 2V
- 5.8uA Sec_Idle, 2V
- 1.1uA Timer1_Osc, 2V
- 0.1uA Sleep, 2V
- 2倍速启动
- **BOR 17uA**

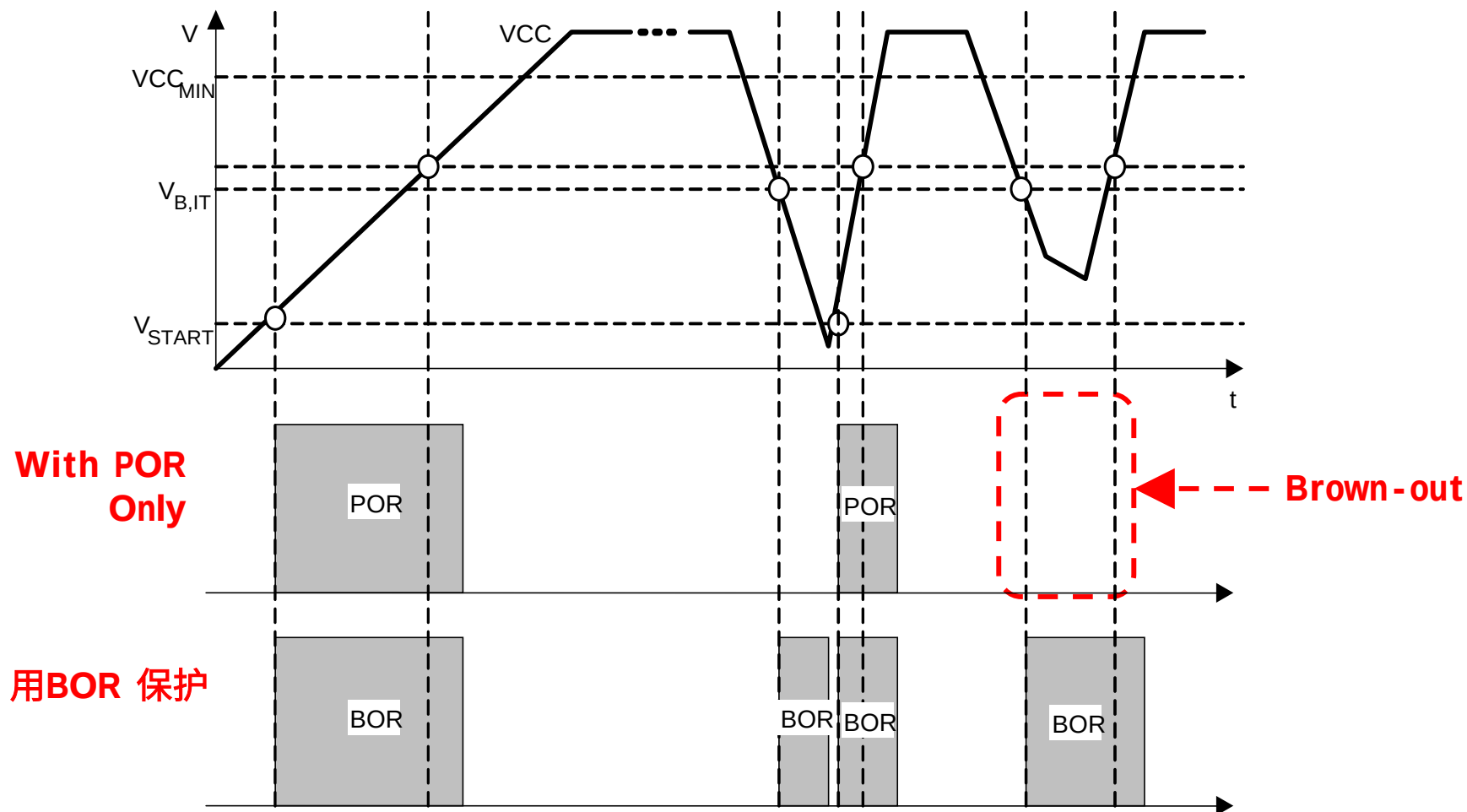
Is Brown-out protection important?

你想设计常用电流还是最大电流?

"LVLP"				
Stop 3	VCC	Typical	Max	Temp
	3	675 nA	4.3 uA	55C
	2	500 nA	7.2 uA	70C
			17.0 uA	85C
输入漏电流			3.5 uA	55C
RTI adder			6.2 uA	70C
LVI adder			15.0 uA	85C
		25 nA	1uA	
		300 nA		
		70 uA		

没有为管脚设计低功耗

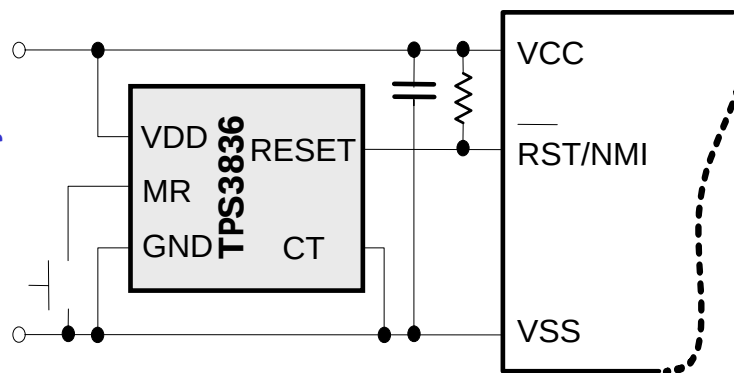
买到产品 - *超低功耗的* BOR 保护



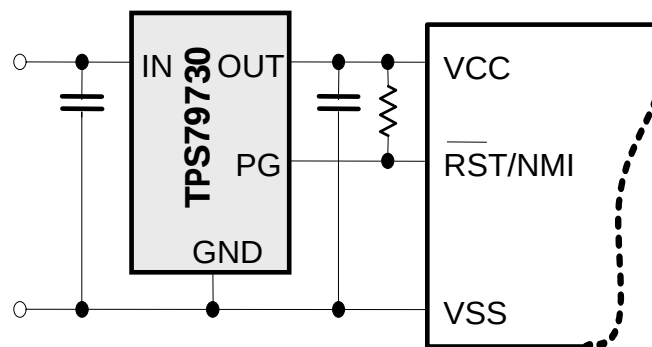
外部供压保护

应用报告:
SLVA091

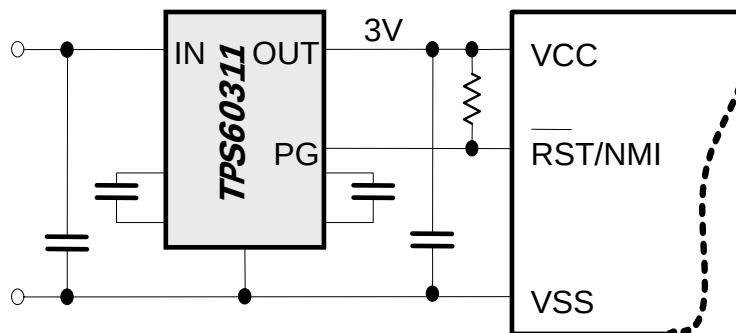
220nA SVS
数据手册 **SGLS151**



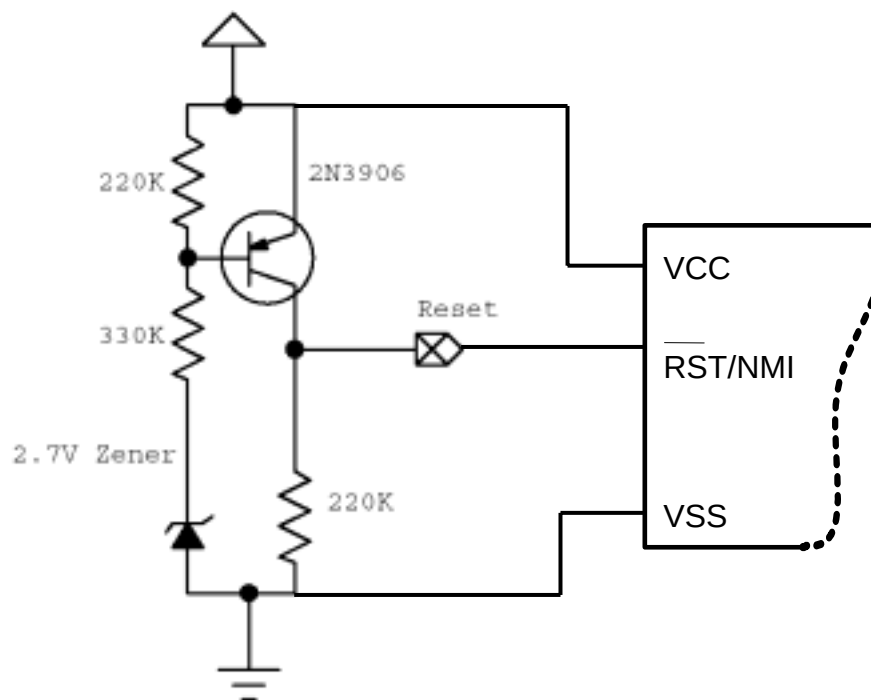
1.2uA LDO w/ 优良电源电路
数据手册 **SLVS332**



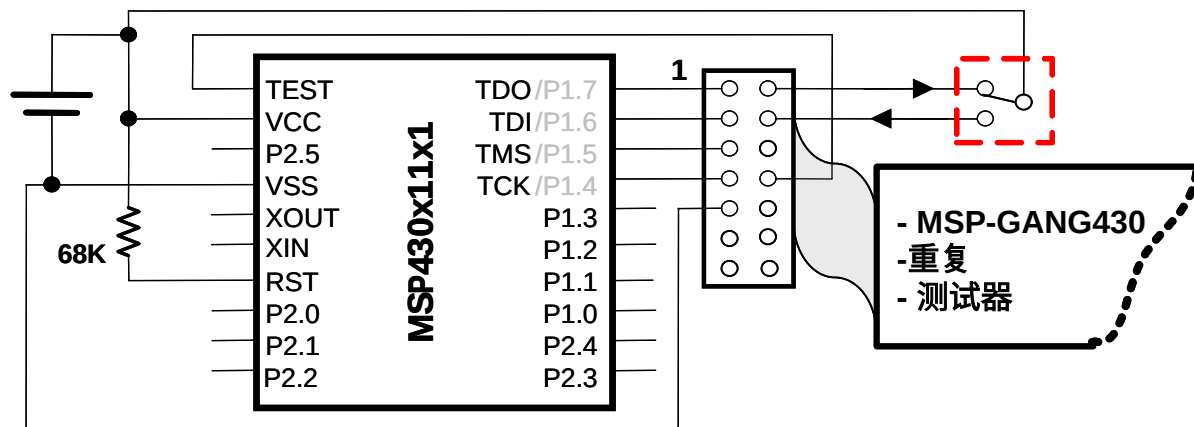
2uA DC/DC w/ 优良电源电路
数据手册 **SLVS362**



30uA低耗 Brown out 电路

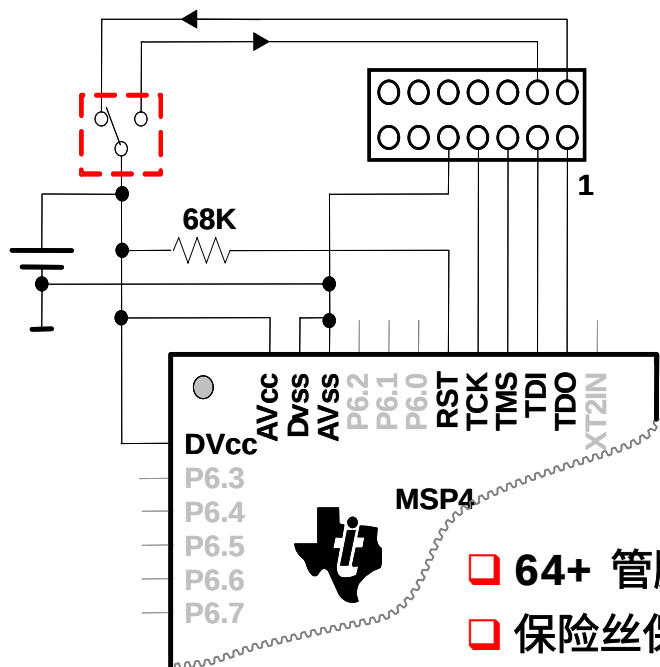


JTAG 支持仿真/编程



- ☐ 20/28 管脚器件
- ☐ Test引脚使能JTAG 信号
- ☐ 保险丝保护存取

Vcc可以使用JTAG连接器引脚的电源，或者将来自JTAG口4脚的目标板电源作为反馈调节电平阈值

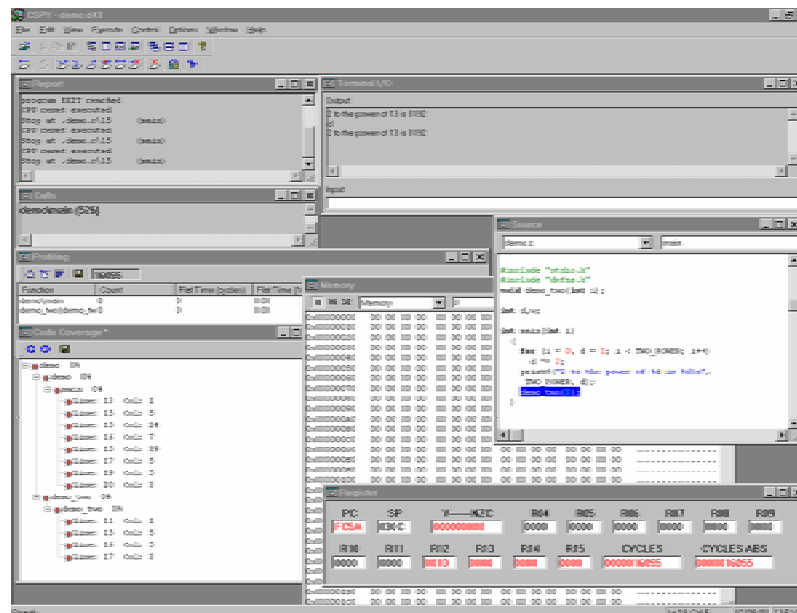


- ☐ 64+ 管脚器件
- ☐ 保险丝保护存取

现在准备开始了！

MSP430-FET \$99 IDE

- ❑ 实时嵌入式仿真
- ❑ JTAG 界面
- ❑ 目标板, 两个器件
- ❑ 无限制的Assembler / linker
- ❑ IAR 4kB_C-编译器
- ❑ CD-ROM / 全部文档



TI IDE

MSP-FET430X110
MSP-FET430P120
MSP-FET430P140
MSP-FET430P410
MSP-FET430P440

系列产品

MSP430x11x
MSP430x11x/12x(2)
MSP430x13x/14x/15x/16x
MSP430x41x/42x
MSP430x43x/44x

Project: FET_16X, FET_42X

在线 JTAG 编程器

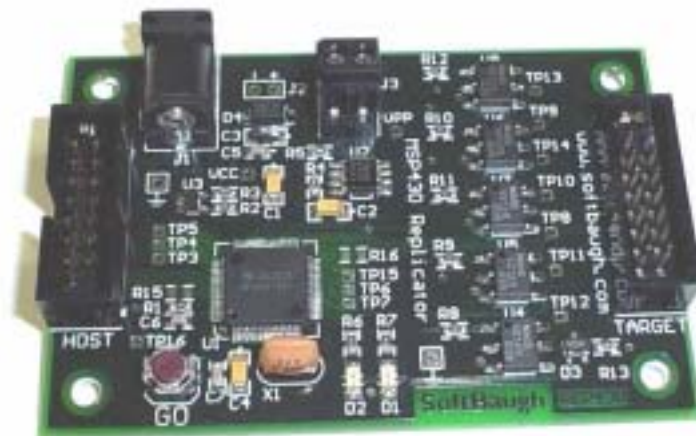
应用报告:
SLAA149

- ❑ MSP-GANG430
- ❑ 同时对8个设备进行编程
- ❑ 通过RS232可以进行单个或GUI编程控制
- ❑ ~7kB/s (60kB in 9s)
- ❑ 安全保护熔丝编程
- ❑ \$199 USD



用户指南:
SLAU101

- ❑ 自制编程器或从第三方购买
- ❑ 擦除、编程、校验
~8 kB in 1.5 sec (48 KB in 8 sec)
- ❑ 最大编程大小: ~57 KB



产品核对

- ☐ 标准定义容易调试
- ☐ Vcc是自由的脉冲干扰在设备的详细说明
- ☐ ESD 设计知道已经提供.
- ☐ 适当的设置看门狗 .
- ☐ D_{VCC} 和 A_{VCC} 永远保持相同的电压.
- ☐ 闲置的 I/O 管脚按不使用的IO进行设置.
- ☐ 所有 JTAG 管脚不连结.
- ☐ XTAL旁边没有其它信号经过，并能敷地.
- ☐ 如果用 HF-XTAL,依次接通。 .
- ☐ 在 '4xx 器件上, the OSC_CAP 电容被使能.
- ☐ USART应该适当的设置，且SWRST 触发.